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Paper Code : 21309
F–409

B. C. A. (Second Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N—204

PROGRAMMING IN C

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) List the various types of operators available in 'C'. Write the hierarchical order of evaluation of these. 7

   (b) Write a function which accepts an array of size *n* containing integer values and returns the average of all values. Call the function from main program. 7

2. (a) What is an array ? Show different methods to declare and initialize one-dimensional array. Draw how an integer array of 6 elements having base address 5001 is stored in memory. Explain actual storage of two dimensional array *d* [3] [4] in the memory with diagram. 7

   (b) Write and explain the action of 'while' statement. Develop a program in 'C' language to compute the average of every third integer number lying between 1 and 100. Include appropriate documentation. 2, 5

3. (a) Explain with the help of an example and flow-charts the action of break statement and the continue statement. 7

   (b) What are the disadvantages of vested if else statement ? Give an example of nesting of loops. 5, 2

4. (a) Write a 'C' program to find out sum of diagonal elements of a matrix. 7

   (b) Input two different arrays of integers. Write a 'C' program using pointer variables to merge them into third array of sufficient size. 7

5. (a) How does passing an array as an argument to a function differ from call by value ? Is it legal for a function f 1 and f 2, which then calls f 1 ? Justify your answer. 5, 2

   (b) Define recursion. Write a C program to find out factorial of a given number using function recursion. 2, 5

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6. (a) What is dynamic memory allocation ? Mention four functions used for dynamic memory manipulation. Give an example of dynamic memory allocation. 1, 4, 2

   (b) What do you mean by macros ? Write a macro that clears an array to zero. 3, 4

7. (a) Give the main advantages of storing data as a file. Describe various ways in which data files can be categorized in 'C'. Illustrate by examples. 1, 4, 2

   (b) What do you mean by a pointer variable ? Write a function in 'C', using pointers for the array of elements, for sorting the elements. 1, 6

8. (a) What are the various storage classes available in 'C' ? Explain and give *one* example of each. 7

   (b) Write a function to accept two strings as arguments. The function should return the index of the second storing in the first string, if it exists, and –1 otherwise. Don't use any library routines related to strings. 7

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